

The Standard Trade Model

ECON 308 – International Economic Relations Chapter 6

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The Standard Model: A Unified Framework

- The Ricardian model has a straight-line PPF (one factor).
- The H-O model has a bowed-out PPF (two factors).
- The **standard trade model** works with any bowed-out PPF — it nests both as special cases and allows us to answer:
 1. How do changes in the *terms of trade* affect welfare?
 2. How does *economic growth* affect trade and welfare?
 3. When can a country benefit from imposing a tariff?

Why This Matters

The standard model provides the clearest framework for understanding how changes in *world prices* affect national welfare — a key question for Canada as a commodity exporter.

Learning Objectives

1. Define the terms of trade and explain why they determine welfare.
2. Use the standard model to analyse growth and its effects on the ToT.
3. Distinguish export-biased from import-biased growth.
4. Explain the conditions for immiserising growth.
5. Analyse how tariffs and export subsidies affect the terms of trade and welfare.
6. Apply these concepts to Canadian trade policy debates.

The Production Possibility Frontier (Bowed-Out)

Why bowed-out?

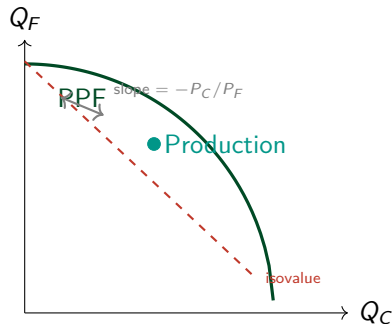
With multiple factors, increasing production of cloth requires pulling factors that are progressively less suited for it (more capital-intensive workers, land not suited for textile mills).

Opportunity cost of cloth rises as cloth output rises.

Production equilibrium: the economy produces where the **isovalue line** is tangent to the PPF:

$$P_C \cdot Q_C + P_F \cdot Q_F = \text{maximised}$$

The slope of the isovalue line = $-P_C/P_F$.



Consumption Choices and Social Indifference Curves

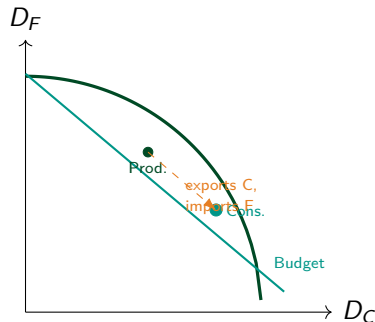
Consumption equilibrium: maximise social welfare $U(D_C, D_F)$ subject to the budget constraint:

$$P_C \cdot D_C + P_F \cdot D_F = P_C \cdot Q_C + P_F \cdot Q_F$$

where D_C, D_F are consumption of cloth and food.

Key: under trade, the budget line has slope $-P_C/P_F$ (world prices) and passes through the *production* point.

Consumption can be **anywhere on the budget line** — not just on the PPF.



Terms of Trade: Definition and Welfare Link

Definition

The **terms of trade (ToT)** for a country = price of its *exports* divided by price of its *imports*:

$$\text{ToT} = \frac{P_{\text{export}}}{P_{\text{import}}}$$

For Home (cloth exporter): $\text{ToT} = P_C/P_F$.

Welfare and the ToT:

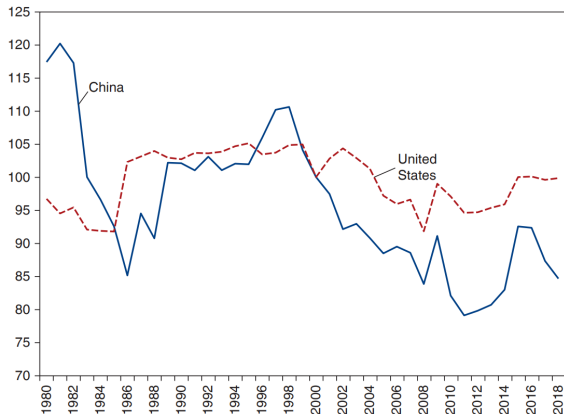
- ToT $\uparrow$ (export prices rise): budget line rotates outward $\Rightarrow$ higher CIC $\Rightarrow$ welfare **rises**.
- ToT $\downarrow$ (import prices rise): budget line rotates inward $\Rightarrow$ lower CIC $\Rightarrow$ welfare **falls**.
- The ToT is the single most important price variable for a small open economy's welfare.

Canada's ToT 1990–2020

Oil price: $\sim \$20$ (1990) $\rightarrow$ $\$110$ (2014) $\rightarrow$ $\$45$ (2016) $\rightarrow$ $\$85$ (2023).

Canada's ToT closely tracks oil prices. The 1990–2014 improvement added $\sim 15\%$ to real national income *beyond* real GDP growth.

Figure 6-8: Evolution of US Terms of Trade



Reading Figure 6-8:

- **Vertical axis:** US terms of trade index (export prices / import prices), normalised to 100 in some base year.
- **Long-run trend:** relatively stable with cyclical swings driven by energy prices.
- **Oil shocks:** 1973, 1979 oil price spikes worsened US ToT (US imports oil).
- **Post-2008:** shale revolution reduced US oil imports, improving ToT.
- The US is a large economy — its growth and policies *affect* world prices (and hence its own ToT).

Contrast with Canada

Canada is an oil exporter — its ToT moves *opposite* to the US when oil

World Relative Supply and Demand

World Relative Supply (RS): upward-sloping.
Higher P_C/P_F induces both countries to shift production toward cloth $\Rightarrow$ world relative supply of cloth rises.

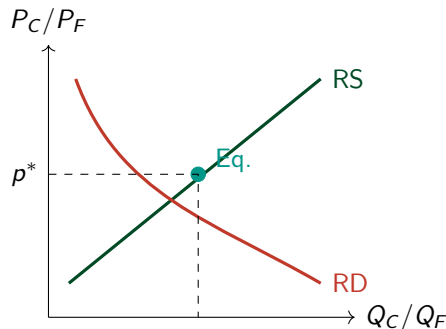
World Relative Demand (RD):

downward-sloping.

Higher P_C/P_F induces consumers worldwide to substitute toward food $\Rightarrow$ relative demand for cloth falls.

World equilibrium: $RS = RD$ determines $p^* = P_C^*/P_F^*$.

Shifts in either curve change the equilibrium ToT.



Economic Growth: Export-Biased vs. Import-Biased

Growth Bias

Export-biased growth: expands production of the export good (cloth) more than the import good (food) at constant prices. RS shifts right $\Rightarrow$ ToT **deteriorates**.

Import-biased growth: expands production of the import good more, or raises demand for imports (biased consumption). ToT also **deteriorates**.

Export-biased growth (Rybczynski):

- Growth in factor used intensively in export good.
- More cloth produced at given prices.
- World price of cloth falls: ToT worsen.

China's Impact on Canada

China's rapid growth (import-biased from Canada's perspective) *increased* demand for Canadian commodities. Canada's ToT improved 2002–2014 as China bought more oil, potash, lumber, and copper. Canada benefited from China's growth.

Immiserising Growth: When Growth Makes You Poorer

Bhagwati (1958): Immiserising Growth

A country can be made **worse off by economic growth** if:

1. Growth is *strongly export-biased* (Rybczynski).
2. The country is *large* enough to affect world prices.
3. World demand for its export is *price-inelastic*.

Then: ToT deterioration *exceeds* the direct production gain $\Rightarrow$ net welfare loss.

Conditions for immiserising growth:

- Large-country effects: more growth worsens ToT more.
- Inelastic demand: price falls a lot when supply rises.
- Strong bias: growth concentrated in export sector.

Canadian Example

BC *dramatically* increases timber harvesting. Global lumber price falls sharply. Revenue may actually *fall* even as output rises.

This is why sustainable forestry quotas

Tariffs and the Terms of Trade

Large-country tariff on food imports:

1. Tariff reduces Home's demand for food imports.
2. World price of food **falls** (foreign exporters sell less).
3. Home's ToT improve: each unit of cloth now buys more food.
4. But: tariff creates domestic deadweight losses.

Net welfare effect:

ToT gain — domestic efficiency loss.

If country is large enough and ToT gain dominates: **optimal tariff** is positive.

The Trade War Risk

The optimal tariff argument is valid for one country in isolation. But if all countries impose optimal tariffs, a *trade war* erupts. The Nash equilibrium (mutual protection) leaves *all countries worse off* than mutual free trade.

This is the prisoner's dilemma of trade policy — solved by GATT/WTO commitments.

Export Subsidies and the Terms of Trade

Export subsidy on cloth:

- Government pays domestic cloth producers per unit exported.
- More cloth flows to world market $\Rightarrow$ world price of cloth falls.
- Home's ToT **deteriorate** (export price falls).
- **Result:** Home consumers pay more for cloth; Home government pays the subsidy; ToT worsen.

Export subsidies are worse than tariffs for the subsidising country: both create domestic distortions but subsidies *also* worsen the ToT.

EU Agricultural Export Subsidies

The EU's Common Agricultural Policy paid billions in export subsidies on wheat and dairy through the 1990s.

This dumped cheap food on world markets, hurting farmers in developing countries and worsening EU ToT.

Under WTO pressure (Doha Round), EU agreed to phase out export subsidies. Fully eliminated by 2013.

Key Takeaways

1. The standard trade model uses bowed-out PPFs and social indifference curves to analyse welfare.
2. **Terms of trade** are the key welfare variable: ToT improvement raises welfare; deterioration lowers it.
3. **Export-biased growth** worsens a country's ToT; **import-biased growth** in trading partners can improve your ToT.
4. **Immiserising growth** is possible if growth is export-biased, the country is large, and world demand is inelastic.
5. **Large-country tariff** improves ToT but invites retaliation — trade wars hurt all.
6. **Export subsidies** are even worse: they simultaneously distort the domestic economy and worsen the ToT.

Next: Chapter 7 — Economies of Scale and Trade

Why do similar countries trade similar goods? Intra-industry trade and the Krugman model.